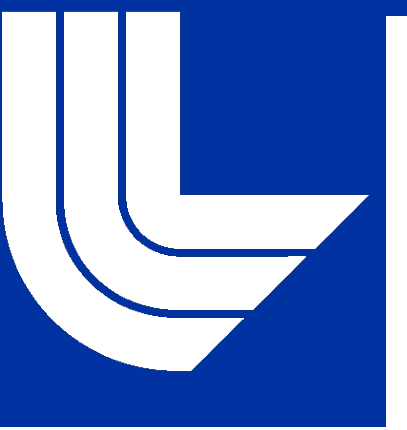




Explicit-Implicit-Null Time Stepping: Turning Stiff, Nonlinear Anisotropic Diffusion into Fast, Linear Solves

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The Explicit-Implicit-Null (EIN) time integrator is a novel IMEX scheme developed to linearly treat stiff, nonlinear spatial derivative terms. This strength bypasses time-step size limitations and nonlinear costs, common in traditional temporal solvers. We implement EIN within the MFEM library and verify its performance up to third-order accuracy on nonlinear diffusion problems. Moreover, we find that higher-order variants scale down stiffness, demonstrating the value of high-order methods to performance scaling in scientific computing.

MOTIVATION: Temporal Bottlenecks

High-fidelity physics solvers are often throttled by extreme stiffness from leading spatial derivatives:

- **CFL Restrictions:** Explicit methods are severely restricted by parabolic or worse time-step limits.
- **Nonlinear Costs:** Standard IMEX methods must solve large, nonlinear, and expensive systems when physics are self-dependent.

At LLNL, the **Modular Finite Element Methods (MFEM)** codebase needs novel timesteppers that address these common temporal bottlenecks.

Objective: Extend the EIN approach to enable stable, linear-solve-only time integration for nonlinear physics contexts in MFEM.

Model Problem: Nonlinear Anisotropic Diffusion

Many systems exhibit temperature evolution as directional diffusion:

$$T_t = \nabla \cdot (\mathbf{D}(T) \nabla T).$$

Diffusion tensor $\mathbf{D}$ depends on magnetic field direction $\mathbf{b} = (\cos \alpha, \sin \alpha)$, for α degrees of misalignment, and its directional diffusion rates

- **Parallel Diffusion:** $D_{\parallel}(T) \sim T^{2.5}$
- **Perpendicular Diffusion:** $D_{\perp}(T) \sim \text{constant}$.

Misalignment of magnetic field from grid, nonlinear scaling, and extreme anisotropy can all induce extreme stiffness.

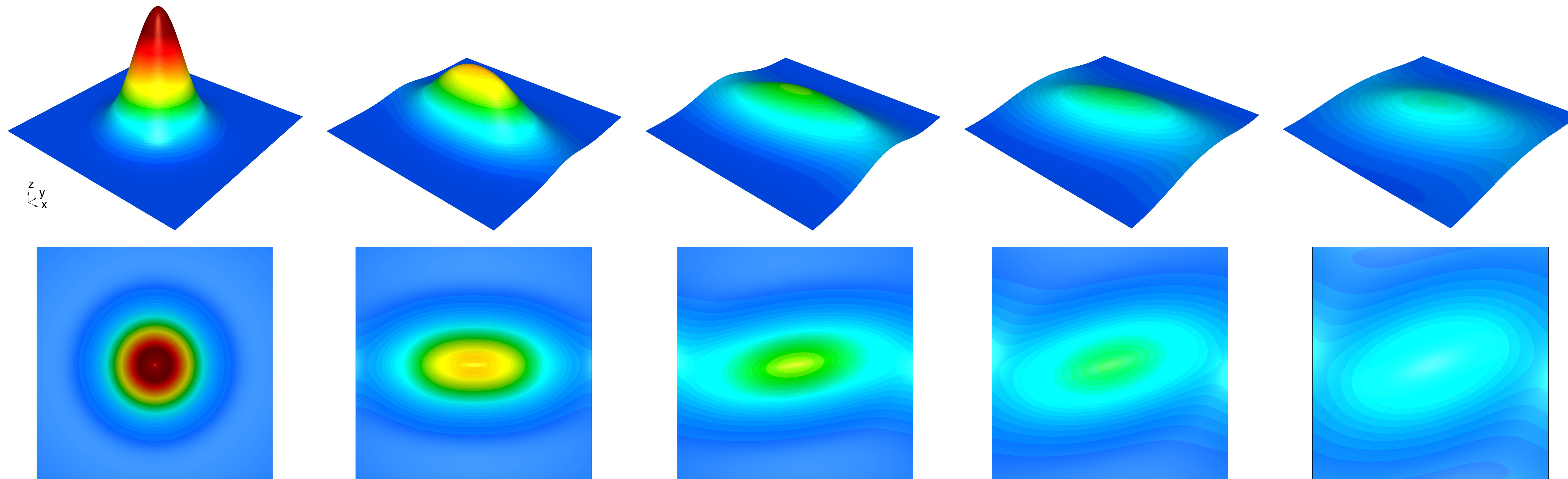


Figure 1. Nonlinear anisotropic diffusion of a Gaussian, with varying α . (Top: surface, Bottom: contour) Strong anisotropy develops a ridge in parallel direction, rotating as α changes from 0° to 90° .

THE EIN FRAMEWORK: Nonlinear $\rightarrow$ Linear

For method-of-lines on general convection-diffusion

$$u_t = F(u, t),$$

stiff spatial derivatives in F are often **nonlinear**, requiring expensive solves.

EIN isolates stiffness by adding/subtracting a “null” linear approximation G :

$$u_t = \underbrace{F(u, t) - G(u, t)}_{\text{Explicit (Non-stiff)}} + \underbrace{G(u, t)}_{\text{Implicit (Stiff, Linear)}}$$

Correct choice of G gives **unconditional linear stability** in high-order schemes for convection-diffusion like the compressible Navier-Stokes.

For **anisotropic diffusion**, we apply EIN as

$$T_t = \underbrace{\nabla \cdot (\mathbf{D}(T) \nabla T)}_{\text{Explicit}} - \underbrace{G(T, t)}_{\text{Implicit}} + \underbrace{G(T, t)}_{\text{Implicit}}$$

and choose $G(T, t)$ as appropriately large **linear anisotropic diffusion**, via the same parallel and perpendicular directions:

$$G(T, t) = a_0 (\max(D_{\parallel}) \nabla \cdot (\mathbf{b} \mathbf{b}^T \nabla T) + \max(D_{\perp}) \nabla \cdot (\mathbf{b}_{\perp} \mathbf{b}_{\perp}^T \nabla T)).$$

This encodes the diffusion and anisotropy, while still **linear** – allowing fast solvers instead of nonlinear iterations.

Linear stability analysis on IMEX schemes informs of the stability factor a_0 needed to ensure **unconditionally stability** of the EIN application.

RESULTS: Isotropic & Anisotropic

- **Numerical Convergence & Accuracy Verification:** optimal orders are achieved for IMEX-EIN timesteppers of orders 1, 2, and 3.

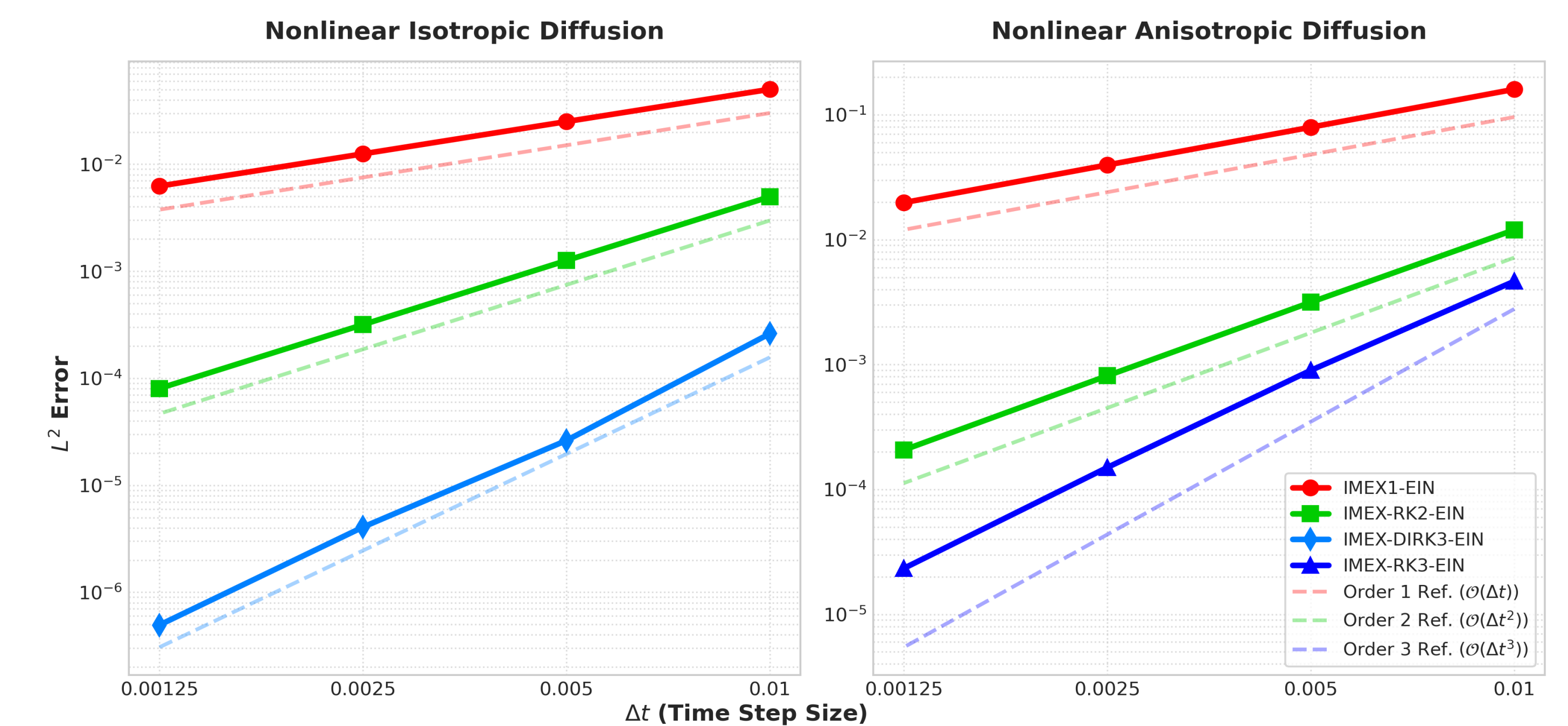


Figure 2. Error convergence for 2D problems. Left: nonlinear form of $a(T) = \kappa + \eta T$, Right: nonlinear form of $D_{\parallel}(T) = T^{2.5}$ with time-varying α .

- **High-Order Performance:** for a cheap, simple preconditioner, high-order schemes better condition systems and decrease error magnitude.

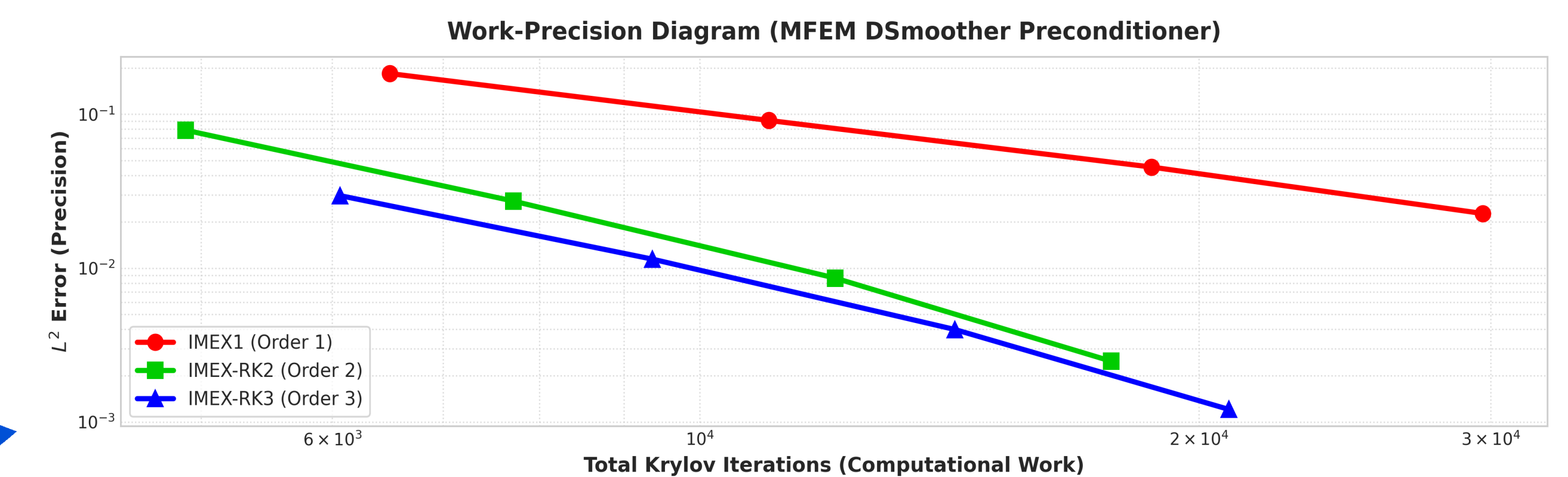


Figure 3. Work-precision trade-off among IMEX-EIN schemes for 2D anisotropic diffusion with nonlinear form of $D_{\parallel}(T) = T^{2.5}$ and time-varying α .

CONCLUSIONS

The EIN framework is a promising building block for complicated physics:

- **Novel:** transforms stiff, nonlinear problems into stable, linear ones.
- **Robustness:** handles misalignment and various nonlinearities.
- **High-Order Value:** demonstrates how high-order methods are vital to computational speedups in scientific computing.

Future Work: coupling anisotropic thermal diffusion within larger physical systems in MFEM.

[1] Anderson et al., *Comput. Math. Appl.* (2021). [2] Ascher et al., *App. Num. Math.* (1997). [3] Calvo et al., *App. Num. Math.* (2001). [4] Chacón et al., *J. Comp. Phys.* (2025). [5] Tan et al., *J. Comp. Phys.* (2025). [6] Tan et al., *J. Comp. Phys.* (2022). [7] van Es et al., *J. Comp. Phys.* (2014). [8] Wang et al., *J. Sci. Comp.* (2020).